Supporting Information: "Have Molecular Foundation Models Improved for Free-Energy Sampling?"

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Training Data – Azobenzene

The azobenzene structures were drawn from 12 independent two-dimensional well-tempered metadynamics trajectories: one *cis*- and one *trans*-initiated trajectory for each of MACE-OFF24, MACE-OMOL, MACE-MH1, MACE-POLAR, PET-SPICE, and SO3LR. Before the sampling, each initial structure was optimized using BFGS. The biased trajectories used the CNNC torsion, ϕ, and CNN bending angle, θ, as collective variables. Gaussian hills of height 0.1 eV and widths $\sigma_\phi = \sigma_\theta = 5^\circ$ were deposited every 100 MD steps. Simulations used a timestep of 0.5 fs, a Langevin friction coefficient of 0.1, and a well-tempered bias factor of 10. From every trajectory, 200 saved configurations were selected uniformly at random using a fixed random seed, yielding 2,400 molecular structures in total. No additional selection based on CV position, energy, or structural similarity was applied. All selected structures were evaluated by gas-phase single-point calculations at the ωB97M-V/def2-TZVPD level of theory. The resulting dataset was randomly shuffled and split on a structure-wise basis into 1,920 training, 240 validation, and 240 test structures (80:10:10). Figure 1 shows the locations of these structures in CV space.

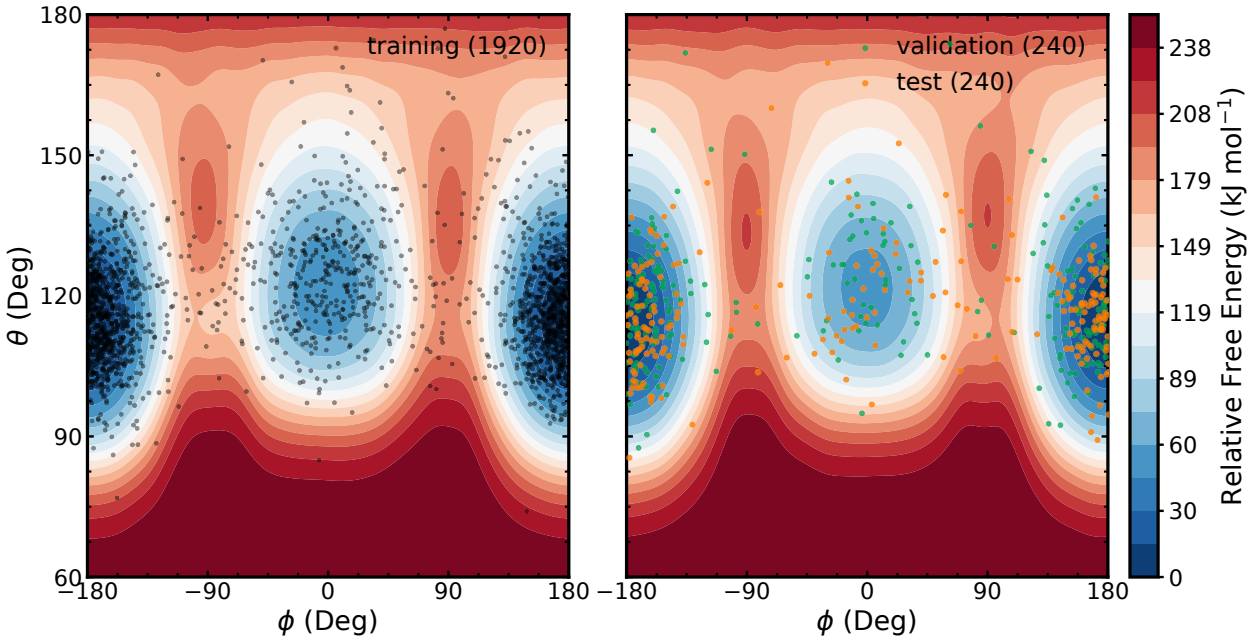


Figure 1: Distribution of the randomly selected azobenzene structures in the CNN bending-angle (θ) and CNNC torsional-angle (ϕ) space. Training structures are shown over the *cis*-initiated fine-tuned MACE-MH1 free-energy surface, while validation and test structures are shown over the corresponding *trans*-initiated surface. The free-energy surfaces are used solely as a common visual reference for the data distribution.

Combination, Symmetrization, and Pathway Analysis – Azobenzene

For each model, separate free-energy surfaces were reconstructed from the *cis*- and *trans*-initialized wall-restrained metadynamics simulations. Since the absolute free-energy offset of each reconstruction is arbitrary, the two surfaces were first aligned independently to their respective global minima,

$$F_i^0(\theta,\phi) = F_i(\theta,\phi) - \min_{\theta,\phi} F_i(\theta,\phi), \qquad i \in \{\mathrm{cis}, \mathrm{trans}\}. \tag{1}$$

They were then combined by a pointwise arithmetic mean,

$$F_{\mathrm{comb}}(\theta,\phi) = \frac{1}{2}\left[F_{\mathrm{cis}}^0(\theta,\phi) + F_{\mathrm{trans}}^0(\theta,\phi)\right]. \tag{2}$$

Finally, the combined surface was symmetrized with respect to torsional inversion,

$$F_{\mathrm{sym}}(\theta,\phi) = \frac{1}{2}\left[F_{\mathrm{comb}}(\theta,\phi) + F_{\mathrm{comb}}(\theta,-\phi)\right] - \min_{\theta,\phi}\frac{1}{2}\left[F_{\mathrm{comb}}(\theta,\phi) + F_{\mathrm{comb}}(\theta,-\phi)\right]. \quad (3)$$

Thus, the final symmetrized surface has a zero global minimum. This procedure averages independently reconstructed FESs; it does not pool or reweight the underlying bias histories.

Figure 2 shows the individual, unsymmetrized FESs for all eight models before these operations. The simulations included smooth restraining walls on both adjacent CNN angles. The wall energy was zero between 45° and 170°, rose quadratically to 5 eV between 35° and 45°, and likewise rose to 5 eV between 170° and 180°. Consequently, the FES close to $\theta = 180°$, and especially the upper part of the high-angle pathway corridor, is affected by the restraining potential.

Two constrained pathways were evaluated for each surface. The low-angle route was obtained as a smooth minimum-cost path within the $\theta = 105°$–$135°$ corridor. The high-angle route was obtained using a minimax criterion: among smooth paths connecting the low-angle *cis* and *trans* regions, the selected path minimizes the largest FES value encountered while being required to cross the region

$$50° \le \phi \le 130°, \qquad 155° \le \theta \le 175°. \quad (4)$$

The low- and high-angle pathways should therefore be interpreted as conditional route comparisons, rather than unconstrained minimum-free-energy paths.

The pathways obtained with the foundation SO3LR model are not physically interpretable, because its individual FESs do not resolve distinct localized *cis* and *trans* basins in the relevant region of CV space.

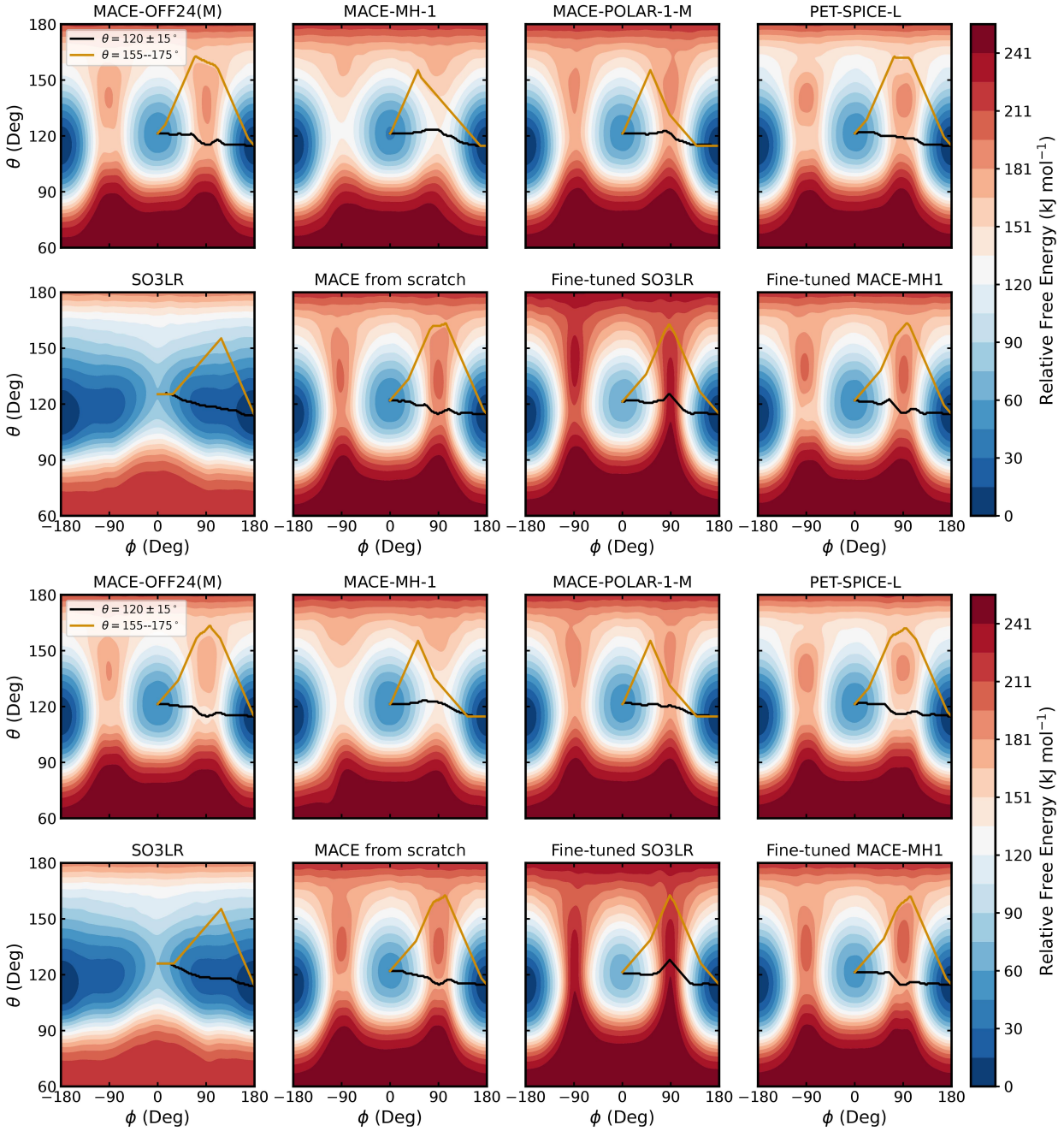


Figure 2: Unsymmetrized FESs from *cis*-initialized (top) and *trans*-initialized (bottom) metadynamics simulations. Black and amber lines denote the low-angle and constrained high-angle pathways, respectively. These surfaces are aligned, averaged, and torsionally symmetrized for the combined FES analysis.

Replicated AMBER Reference Simulations

AMBER ff14SB/TIP3P vs AMBER ff19SB/OPC

The AMBER reference free-energy surfaces for solvated Ac–Ala_3–NMe were obtained from four independent two-dimensional well-tempered metadynamics simulations using AMBER ff14SB/TIP3P and AMBER ff19SB/OPC. The ff19SB/OPC simulations used OpenMM[1] to evaluate the force field, including the OPC virtual sites, while biASE applied the metadynamics bias to the real atoms. The in the manuscript shown FES plots of AMBER ff14SB/TIP3P and AMBER ff19SB/OPC are to zero shifted averages of these replica.[2–5]

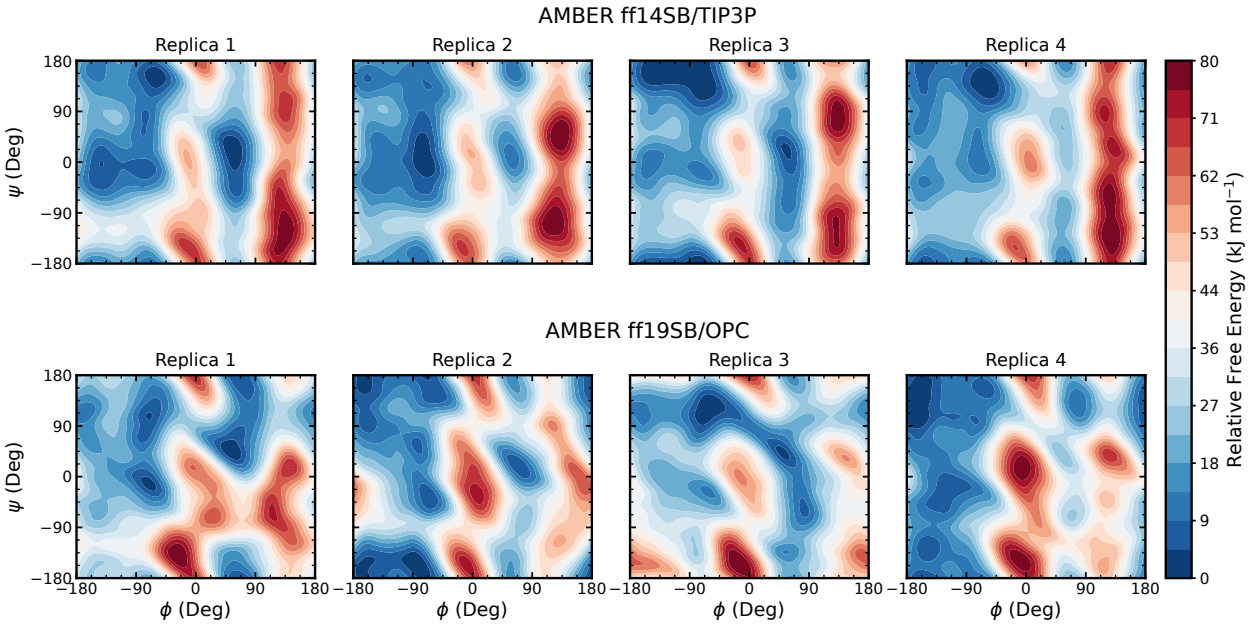


Figure 3: Free-energy surfaces of solvated Ac–Ala_3–NMe obtained from four independent 1 ns well-tempered metadynamics replicas for AMBER ff14SB/TIP3P (top row) and AMBER ff19SB/OPC (bottom row).

AMBER ff14SB/TIP3P Replica Variation: biASE and OpenMM

To validate the biASE-based metadynamics implementation, four independent 1 ns well-tempered metadynamics simulations of solvated Ac–Ala_3–NMe were performed with AMBER ff14SB/TIP3P using biASE and, separately, OpenMM's native metadynamics imple-

mentation.

Figure 4 shows that the reconstructed FESs vary appreciably between individual 1 ns replicas for both implementations.

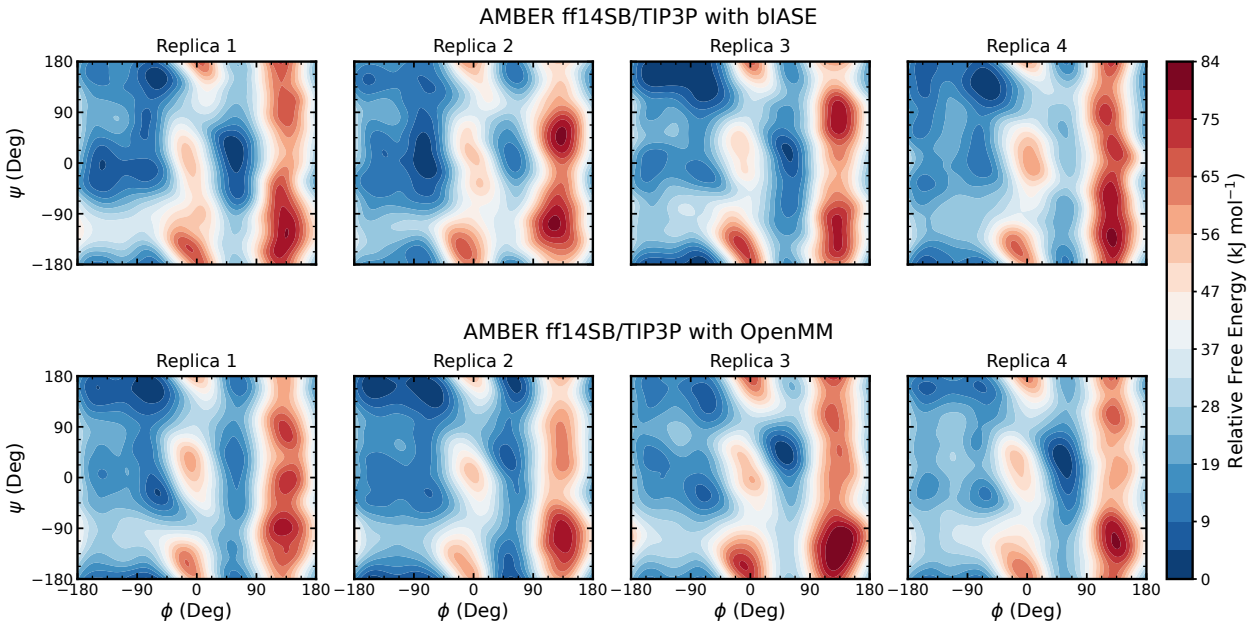


Figure 4: Individual free-energy surfaces from four independent 1 ns AMBER ff14SB/TIP3P simulations of solvated Ac–Ala_3–NMe. The top row was generated using biASE metadynamics and the bottom row using OpenMM's native metadynamics implementation. Each panel is reconstructed from the corresponding deposited bias and shifted to a zero minimum.

References

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